understanding why an application behaves in a certain way. You can monitor metrics and set alerts or notifications for them.

Spring Boot Actuator collects application metrics out of the box by leveraging the Micrometer library (https://micrometer.io). Micrometer contains instrumentation code for collecting valuable metrics from common components in a JVM-based application. It provides a vendor-neutral façade so that you can export the metrics collected from Micrometer using different formats, such as Prometheus/Open Metrics, Humio, Datadog, and VMware Tanzu Observability. Just as SLF4J provides a vendor-neutral façade for logging libraries, Micrometer does the same for metrics exporters.

On top of the default Micrometer instrumentation libraries that are configured by Spring Boot, you can import additional instrumentation to collect metrics from specific libraries like Resilience4J or even define your own without vendor lock-in.

The most common format for exporting metrics is the one used by Prometheus, which is “an open-source systems monitoring and alerting toolkit” (https://prometheus.io). Just as Loki aggregates and stores event logs, Prometheus does the same with metrics.

In this section you’ll see how to configure metrics in Spring Boot. Then you’ll use Prometheus to aggregate metrics and Grafana to visualize them in dashboards.

13.3.1 Configuring metrics with Spring Boot Actuator and Micrometer

Spring Boot Actuator auto-configures Micrometer out of the box to collect metrics about a Java application. One way of exposing such metrics is by